Cleaning data: duplicates, blanks and wrong types

Lesson notes · Module 1 · Sample material for the demo

Why cleaning comes first

Every total, average and chart you build is only as good as the rows underneath it. A shop file with a sale typed twice will overstate revenue; a quantity stored as text will be silently ignored by SUM. Ten minutes of checking saves hours of explaining a wrong number to your manager.

The five checks to run on any new dataset

1. Count the rows.
Write down how many rows the file has before you change anything. Count again at the end. The difference is exactly what your cleaning removed.

2. Find duplicates.
Sort by the receipt or transaction number. Two rows with the same number, date and amount are almost always the same sale entered twice. Data > Remove Duplicates does this in one step, but look before you delete.

3. Decide what blanks mean.
A blank quantity can mean zero, unknown or forgotten. Check the source (the receipt book, the clinic register) before you fill or delete. Never type 0 just to make a formula work.

4. Fix numbers stored as text.
Numbers aligned to the left, or a SUM that returns 0, are the giveaway. Use VALUE(), or Data > Text to Columns, to convert the column.

5. Make categories consistent.
"Kigali", "kigali " and "KGL" are three different values to Excel. Use TRIM, PROPER and Find & Replace so each category is spelled one way.

Worked example

A Nyabugogo shop records five sales in a week: RWF 12,000 on Monday, 8,500 on Tuesday, 12,000 on Tuesday (same receipt no. 0412), 9,000 on Wednesday and 20,500 on Friday. The raw total is RWF 62,000. Receipt 0412 appears twice, so one row is removed and the real total is RWF 50,000: a 24% overstatement from a single duplicate.

Your turn

Open the shop sales file from the exercise. Run the five checks, keep a short note of each change and its reason, and compare your row count before and after. You will hand in the cleaned file and the note in the next lesson.